Nutrition · Healthy Habits

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FINAL WORK Nutritional and nutritional intervention health habit change ANALYSIS OF CLINICAL CASES: MANUEL AND CLAUDIA Nutrition work, food education and promotion of physical activity Document content: nutritional assessment, dietary changes, nutritional education and physical activity plan Learn to eat better, move more and consolidate healthy habits to improve quality of life.

FINAL WORK PATIENT 1:

Manuel is a 65-year-old man, single, retired since he was 57.

due to a myocardial infarction. The family doctor has referred to a dietitian nutritionist for poor results biochemical tests in the last two tests and due to his weight gain in the last year. Always has a tendency towards obesity, measures 170 cm and his current weight is 91.7 kg. Since the heart attack, he has been taking medication for hypercholesterolemia and hypertension, as well as Sintrom (anticoagulant drug). In your His last blood test had increased his blood levels.

LDL cholesterol and decreased HDL, so it was necessary adjust the medication again. He has never practiced physical exercise because it seems like a waste of time. Although, forced by the situation, for a month He is going to walk 2 km in the afternoons.

Below is what your intake for a day could be any weekday:

Time Place Food/Drink 9:30 a.m. Coffee with whole milk and sugar 11:15 a.m. Bar next to your house Potato omelette skewer, one cane 2:15 p.m. Vegetable puree or salad or legumes with chorizo, meat (pork rib, chicken or steak beef) with peppers or other vegetables, a coffee cut with sugar, two glasses of red wine 7:30 p.m. Bar (with friends) Two beers, nuts or potatoes 9:30 p.m. Cured cheese sandwich or chorizo or ham or ham tortilla, a glass of wine

FINAL WORK

1. Review of the nutritional and biochemical situation Manuel has grade I obesity (BMI ≈ 31.7 kg/m²), with history of myocardial infarction, poorly controlled dyslipidemia (↑ LDL, ↓ HDL), hypertension and anticoagulant treatment with Sintrom.

It has a dietary pattern that is hypercaloric, rich in fat.

saturated, alcohol, simple sugars and processed products, with low presence of fruit, various vegetables, fish and cereals integral.

From the biochemical and clinical point of view, their diet and Current habits favor:

• Worsening of the lipid profile.

• Increased cardiovascular risk.

• Weight increase and insulin resistance.

• Potential interferences with medication (alcohol and Syntrom).

• Risk of hidden sarcopenia due to poor protein quality and little physical activity. The daily walk that they force him to take is a positive first step, but clearly insufficient to compensate for the rest of the style of life. If you do not change your habits, you are at risk of a new event.

cardiovascular, uncontrolled INR and progression of obesity is elevated.

2. Recommended changes in diet and routine In Food you should:

• Clearly reduce daily Alcohol (ideally eliminate it or limit it to specific occasions).

• Reduce or preferably eliminate sausages, chorizo, cured cheese and fatty meats.

• Eliminate added Sugar (coffee, drinks).

• Get out of the habit of Fritters and skewers at bars.

FINAL WORK It is recommended to increase consumption of:

• Vegetables and vegetables for lunch and dinner.

• Daily fruit (2 – 3 pieces).

• Fish, especially blue, 2 – 3 times/week.

• Legumes without sausages.

• Extra virgin olive oil in moderate quantity.

You should also change whole dairy products for semi-skimmed or skimmed and to adjust portions and schedules, avoiding type dinners fatty snack.

Create a Routine:

• You should maintain your daily walk and progress from 2 km at 30 – 45 min at a brisk pace, introduce strength exercises gentle 2 days/week (very important at his age).

• Reduce social time linked exclusively to alcohol (which the bar is not the only social plan).

• Sleep and eat at more regular times.

3. Example of a complete menu for one day Moment Food/Drinks Breakfast Coffee with semi-skimmed milk without sugar, whole wheat toast with tomato and olive oil extra virgin, a piece of fruit Mid-morning Natural skimmed yogurt or a small handful unsalted nuts, water Food Stewed lentils with vegetables (without chorizo), baked fish fillet with vegetables sautéed, whole wheat bread (small amount), fruit, water (without wine) Snack Fruit or natural yogurt, infusion or coffee without sugar Dinner Complete salad with lettuce, tomato, carrot, boiled egg and olive oil, chicken breast or white fish griddle, small whole wheat bread, water

FINAL WORK Expected result if the guideline is followed

• Gradual weight loss (0.5 kg/week).

• Improvement of LDL and increase of HDL.

• Better control of blood pressure.

• More stable INR.

• Lower risk of cardiovascular relapse.

• More energy and better quality of life.

4. Food and Sintrom: what Manuel should know You should know that, with Sintrom, the problem is not that you have to consume vitamin K, but you have to do it more irregular.

Key points to explain to you clearly:

You can eat green leafy vegetables (spinach, chard, broccoli), but always in similar quantities week by week, you should avoid sudden changes in diet without notifying the doctor. alcohol as interferes with INR control, it is best avoided and should not be taken supplements or “natural” products without consulting first.

You should avoid supplements of:

• Vitamin K.

• Ginseng, ginkgo biloba, St. John's wort.

Check INR more frequently if habits change.

FINAL WORK PATIENT 2:

Claudia is a 15-year-old teenager who comes for a consultation referred by his pediatrician due to a diagnosis of overweight and digestive discomfort (frequent constipation), in addition to difficulties concentrating and fatigue. He is 1.68 m tall and weighs 79 kg (BMI ≈ 28).

He lives with his parents, who work outside the home all day, so what you usually eat and eat pre-cooked or quick-cooked foods prepare. Your daily diet is based on foods ultra-processed foods, simple sugars and low fruit consumption, vegetables and fiber. Has not exercised physically for more than a year (she previously attended dance classes), and currently spends more than four hours a day in front of screens (computer, mobile phone, television). He does not have any additional clinical diagnosis and does not take medication. currently, but his pediatrician believes that his diet and lifestyle life should be modified as soon as possible to avoid the development of associated diseases.

Below, what could be an example of a typical day of feeding:

Time Food/Drink Place 7.30 a.m. Packaged juice At home 7.30 a.m. Piece of pastries industrial at the institute 11:00 a.m. Snack like cookies or chocolate bar at the institute 2:30 p.m. Pasta with sauce fried tomato 2:30 p.m. Burger or sausages 2:30 p.m. Sugary soft drink 5:00 p.m. White sliced bread with cocoa cream 5:00 p.m. Chocolate shake 9:30 p.m. Pre-cooked pizza or nuggets or sandwich with French fries 9:30 p.m. Refreshment

FINAL WORK As you did with Manuel, give a brief review of the condition Claudia's nutritional status and the risks associated with her diet current.

Then, answer these questions:

• What changes should you make in your diet and style?

of life to improve your health?

• Could you propose an example of a daily menu adapted to their needs?

• What nutritional education strategies could be applied in your case or family environment?

LIFESTYLE CHANGE TO IMPROVE HEALTH Changes you should introduce into your diet:

Claudia needs to re-educate her eating pattern, “not to get diet” Priority changes:

• Do not skip breakfast, replacing pastries and juices with real foods.

• Reduce ultra-processed foods (pizza, nuggets, pastries, soft drinks) to occasional consumption.

• Eliminate sugary drinks and prioritize water.

• Increase fiber consumption:

• Fruits (2 – 3 pieces/day).

• Vegetables at lunch and dinner.

• Legumes and whole grains.

• Replace refined flours with whole grains.

• Introduce quality proteins: Fish, eggs, legumes, lean meats.

• Establish regular meal times.

• Improve the quality of snacks to avoid peaks in hunger and fatigue.

FINAL WORK These changes will help Claudia:

• Improve constipation.

• Increase satiety.

• Improve energy and concentration.

• Promote weight normalization.

Example of a daily menu adapted to your needs Food Options recommended Accompaniments Drink Breakfast Semi-skimmed milk or natural yogurt, toast whole wheat with olive oil and tomato or turkey piece of fruit Medium tomorrow (institute) Small whole grain sandwich of fresh cheese or turkey, natural yogurt with fruit Food First course: lentils with vegetables or assorted salad; Second course: chicken grilled, baked fish or french omelette Vegetable garnish cooked or sautéed, bread moderate integral, fruit water Snack Whole wheat bread with hummus or avocado, natural yogurt with fruit Dinner Cream of vegetables or sautéed vegetables, fish, egg or breast chicken Small whole wheat bread Water The occasional consumption of soft drinks, pastries and sweets, these should be reserved for special occasions, avoiding its daily consumption to maintain a healthy diet balanced and healthy.

FINAL WORK Nutritional education strategies in the case of Claudia and her family environment We should explain to Claudia, using simple language and directly, the importance of fiber consumption in the prevention and treatment of constipation. Furthermore, it must highlight how a balanced diet positively influences in your daily energy and in your ability to concentrate to key aspects for their academic performance and general well-being.

It is important to avoid messages that may result from guilt, trying at all times to transmit information in a manner constructive and motivating. To facilitate the change of habits, it is I recommend setting small, easy goals.

achievable, allowing Claudia to experience progress without feel overwhelmed. Likewise, their autonomy must be encouraged progressive in the kitchen, encouraging her to participate in the preparation of their own food.

At the family level Parental involvement in weekly menu planning It is essential. It is advisable that there is always food at home visible and accessible healthy foods, thus facilitating positive choices in the day to day. It is also helpful to prepare simple meals with in advance, to guarantee healthy options and avoid resorting to less advisable alternatives in times of hurry.

Key message: Teach Claudia to do a basic reading of Product labels can help you make decisions informed. Also, provide you with snack ideas healthy and quick will make it easier for you to maintain a healthy diet.

balanced even during busiest times. It is It is essential to associate healthy eating with well-being and self-care, avoiding it being perceived as a restriction or prohibition.

FINAL WORK Final key message “It's not about dieting, but about learning to eat better.” now to avoid problems in the future.” PHYSICAL ACTIVITY PLAN Phase 1 · Activation: Progressive pattern of physical activity to teenagers Objective The main objective of this phase is to get out of a sedentary lifestyle without generate rejection towards exercise. It is intended that the movement become part of everyday life, prioritizing the creation of the habit and the enjoyment.

Frequency and duration It is recommended that you perform physical activity between 4 and 5 days a week.

week, with sessions of between 20 and 30 minutes each day. This Frequency allows exercise to be gradually integrated into the routine without being overwhelming.

Type of activity

• Walk at a brisk pace.

• Quiet bike.

• Dancing at home, especially recommended in the case of Claudia.

• Active games such as walking the dog or gentle skating.

strength work Incorporate strength exercises two days a week, dedicating approximately 10 minutes per session. The suggested exercises they are:

• Squats without weight.

• Go up and down stairs.

• Short plank, between 10 and 20 seconds.

FINAL WORK • Exercises using your own body weight.

Screen time management You should take active breaks every 45 – 60 minutes when you are in front of a screen, to reduce sedentary time and promote regular movement.

Key message “Moving a little every day is better than a lot just one day.” day”. This phrase summarizes the importance of perseverance and habit in the practice of physical activity, promoting a more active life and healthy.

PHASE 2 · CONSOLIDATION (Weeks 5 – 8) Objective: The goal of this phase is to improve physical endurance and confidence in the practice of regular physical activity.

Frequency and duration During the weeks that this phase covers, it is recommended that perform physical activity five days a week, with sessions that They last between 30 and 45 minutes each. Being regular is essential to consolidate the progress achieved and promote the creation of healthy habits.

aerobic activity • Walking fast: Ideal to improve ability cardiovascular and keep the body active.

• Bicycle: It is an option that allows you to work on resistance while enjoying the surroundings.

• Structured dance: You can choose to attend classes or follow guided videos, which provides variety and motivation. • Recreational swimming: In addition to being a complete activity, It is gentle on the joints and suitable for all levels.

FINAL WORK Strength exercises You can incorporate strength exercises for two to three days per week, spending between 15 and 20 minutes per session.

These are the recommended exercises:

• Squats: They mainly strengthen the legs and glutes.

• Strides: They allow you to work on balance and leg strength.

lower extremities.

• Push-ups supporting knees: Ideal for strengthening the core progressively higher.

• Plank (20 – 30 seconds): An isometric exercise that activates the core and improves stability.

• Gentle core work: Exercises focused on the area abdominal and lumbar, performed in a controlled manner.

Stretches At the end of each session, it is important to dedicate between five and ten minutes to stretching. This promotes muscle recovery and helps prevent injuries.

Key message “It's not competing, it's progressing.” The importance of focus on personal improvement and gradual advancement, rather than comparison with others.

PHASE 3 · AUTONOMY AND HABIT (Weeks 9 –12) During the third phase, the main objective is to make the exercise physical appearance becomes part of Claudia's identity, consolidating habit and promoting autonomy in sports practice.

Weekly frequency It is advisable to exercise between five and six days a week. week, which allows you to maintain a constant routine and promote the integration of sport into everyday life.

Sessions should last approximately 45 to 50 minutes.

60 minutes. This time is enough to obtain benefits both

FINAL WORK physical as well as mental, and facilitates the progressive adaptation of person to regular exercise.

Distribution of activities

• 3 days of aerobic activity, such as dancing, brisk walking, or ride a bike. With these activities, the cardiovascular capacity and general endurance.

• 2 days of strength exercises combined with weight training core . With this type of training you help strengthen the muscles and improve body stability.

• 1 day of free activity, which can be done with friends or practicing any recreational sport. The objective is enjoy movement and socialize through exercise.

As a main recommendation, it is advisable to return to classes dance if possible, since dance combines aerobic exercise with fun and allows you to reinforce the habit in an enjoyable way.

Key message “The best exercise is the one you want to repeat.” With this phrase summarizes the importance of choosing activities that are motivating and satisfactory, to ensure continuity and enjoyment in sports practice.

x RAMON ALBERTO CURBALAN VEGA PHYSICAL ACTIVITIES TECHNICIAN